Prabhavit Gupta

AI Systems Architect: Generative AI, Multi-Agent Systems & Cloud Deployments



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Summary

AI Systems Architect and Tech Lead with 3 years of hands-on production experience designing, building, and deploying multi-agent systems, enterprise Generative AI architectures, and cloud-native LLM solutions. Proven track record architecting customer-facing AI agents for platforms serving 4,500+ active enterprise users, integrating autonomous tool-calling workflows, vector retrieval pipelines, and robust evaluation suites. Deep practical expertise across modern agentic frameworks, cloud environments (AWS, Azure), and client-facing technical leadership from requirements discovery to production operation. Holder of an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); based in Venice, Italy and available immediately for 100% remote roles.

Technical Expertise

- **Generative AI & Multi-Agent Systems:** Multi-agent architectures, autonomous agent workflows, tool-calling and schema constraints, human-in-the-loop approvals, LangChain/LangGraph patterns, context window optimization, structured outputs.
- **Cloud Platforms & LLM Infrastructure:** AWS (Bedrock, Lambda, API Gateway, S3), Azure (AI Foundry, OpenAI Service), Docker containerization, CI/CD automation with GitHub Actions, Linux server deployments.
- **RAG Pipelines & Data Systems:** Chunking strategies, hybrid vector retrieval (dense + BM25), PostgreSQL with `pgvector`, metadata filtering, reranking, Redis asynchronous message queues.
- **Software Engineering & API Design:** Python (FastAPI, asyncio, pydantic), TypeScript, Node.js, RESTful microservices, webhook orchestration, OpenAPI specifications, automated regression testing with Playwright.
- **Technical Leadership & Delivery:** Solution architecture, client discovery, pre-sales and post-sales technical ownership, stakeholder communication, cross-functional collaboration.
- **Languages & Work Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Designed and operated production customer-facing multi-agent systems and retrieval pipelines for a marketplace platform serving 4,500+ active enterprise users, handling complex inquiries across live catalog APIs.
- Built resilient multi-step reasoning chains with strict schema validation, deterministic guardrails, and human-in-the-loop approval checkpoints, ensuring zero ungrounded actions in production flows.
- Implemented document parsing and ingestion pipelines using LLMs, converting unstructured partner data into validated JSON schemas and expanding live platform inventory.
- Engineered automated evaluation harnesses evaluating system accuracy, grounding, and recall against production transcripts, systematically lowering edge-case error rates.
- Architected and maintained resilient TypeScript/Node.js backend services and PostgreSQL data layers, utilizing asynchronous Redis queues to handle high-throughput query spikes.
- Developed automated integration and end-to-end regression test suites with Playwright, verifying API contracts and preventing system regressions across continuous deployment cycles.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and client-facing delivery of customized web solutions and automated data processing pipelines for European client platforms.
- Built automated data ingestion and transformation services connecting third-party APIs to PostgreSQL databases, maintaining high data freshness and integrity.
- Delivered interactive functional prototypes to validate user flows and technical requirements prior to sprint execution, accelerating cycle times.

Independent Technical Consultant **Aug 2023 - Feb 2024**

AI Automation & Integration Developer *Venice, Italy*

- Built and deployed custom LLM integrations, document retrieval prototypes, and webhook automation workflows for European enterprise clients using Python, TypeScript, and n8n.
- Replaced brittle manual document review workflows with automated LLM-assisted classification and extraction pipelines featuring comprehensive error handling and fallback routing.
- Authored technical architecture specifications, API documentation, and deployment guides to support client engineering teams in operating self-hosted workflows.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.